



Society of Petroleum Engineers

SPE-229820-MS

Seismic Waveform-Driven Characterization Using Full Frequency Inversion (FFI): A Case Study in Enhanced Seismic Reservoir Characterization from the North-West Shelf, Offshore Australia

S. Chatterjee, H. Zhao, and S. Zhang, CNPCRD, Dubai, UAE

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229820-MS](https://doi.org/10.2118/229820-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Accurate characterization of subsurface reservoirs is a core objective in hydrocarbon exploration and development. The quality of seismic reservoir characterization directly influences well placement, field development strategies, and reserve estimation workflows. When seismic inversion is constrained by limited bandwidth and tuning effects, thin beds are merged into composite responses, subtle facies transitions are obscured, and small-scale structural or stratigraphic features are either poorly resolved or completely masked. These limitations have direct operational consequences: missed pay intervals, incorrect mapping of reservoir connectivity, suboptimal placement of production and injection wells, and development plans burdened by unnecessary uncertainty.

Challenges are especially acute in geologically heterogeneous settings where vertical and lateral variations occur at scales close to or below seismic tuning thickness. In these cases, the blending of reflectors caused by limited frequency content in the seismic data can obscure stratigraphic terminations, make thin impermeable baffles invisible, and reduce confidence in both qualitative and quantitative interpretations. The Jurassic Plover Formation in the Poseidon area of the Browse Basin, offshore North-West Australia, is a compelling example. Its interbedded sandstones, siltstones, and shales, deposited in variable-energy environments, form a complex internal architecture that strongly influences reservoir performance. Thin, laterally variable sand bodies and shale drapes exert significant control on fluid flow and sweep efficiency. In such settings, conventional post-stack inversion often reproduces regional impedance trends but fails to resolve the finer-scale architecture that governs production behavior.

Over the past four decades, inversion methods have evolved considerably. Early deterministic acoustic impedance inversions, grounded in the convolutional model, provided a first step in transforming seismic amplitudes into impedance volumes for structural mapping and gross lithologic interpretation. Pre-stack inversion expanded capabilities by integrating amplitude variation with offset (AVO) information to estimate elastic parameters and improve lithology and fluid discrimination. Stochastic and geostatistical inversions sought to inject high-frequency detail through simulation, aiming to honor both seismic data and geologic plausibility. Spectral inversion and thin-bed reflectivity methods attempted to extend usable bandwidth and de-tune seismic responses, offering improvements in vertical resolution. Despite these advances, the vertical resolution achievable with most conventional methods remains fundamentally tied to

the seismic bandwidth and physics of wave propagation (Russell and Hampson, 1991; Partyka et al., 1999). Features thinner than approximately one-quarter of the dominant wavelength are difficult to recover reliably, and any attempt to artificially increase resolution risks introducing noise or geological inconsistency without tight constraints.

Introduction

Accurate characterization of subsurface reservoirs is a core objective in hydrocarbon exploration and development. The quality of seismic reservoir characterization directly influences well placement, field development strategies, and reserve estimation workflows. When seismic inversion is constrained by limited bandwidth and tuning effects, thin beds are merged into composite responses, subtle facies transitions are obscured, and small-scale structural or stratigraphic features are either poorly resolved or completely masked. These limitations have direct operational consequences: missed pay intervals, incorrect mapping of reservoir connectivity, suboptimal placement of production and injection wells, and development plans burdened by unnecessary uncertainty.

Challenges are especially acute in geologically heterogeneous settings where vertical and lateral variations occur at scales close to or below seismic tuning thickness. In these cases, the blending of reflectors caused by limited frequency content in the seismic data can obscure stratigraphic terminations, make thin impermeable baffles invisible, and reduce confidence in both qualitative and quantitative interpretations. The Jurassic Plover Formation in the Poseidon area of the Browse Basin, offshore North-West Australia, is a compelling example. Its interbedded sandstones, siltstones, and shales, deposited in variable-energy environments, form a complex internal architecture that strongly influences reservoir performance. Thin, laterally variable sand bodies and shale drapes exert significant control on fluid flow and sweep efficiency. In such settings, conventional post-stack inversion often reproduces regional impedance trends but fails to resolve the finer-scale architecture that governs production behavior.

Over the past four decades, inversion methods have evolved considerably. Early deterministic acoustic impedance inversions, grounded in the convolutional model, provided a first step in transforming seismic amplitudes into impedance volumes for structural mapping and gross lithologic interpretation. Pre-stack inversion expanded capabilities by integrating amplitude variation with offset (AVO) information to estimate elastic parameters and improve lithology and fluid discrimination. Stochastic and geostatistical inversions sought to inject high-frequency detail through simulation, aiming to honor both seismic data and geologic plausibility. Spectral inversion and thin-bed reflectivity methods attempted to extend usable bandwidth and de-tune seismic responses, offering improvements in vertical resolution. Despite these advances, the vertical resolution achievable with most conventional methods remains fundamentally tied to the seismic bandwidth and physics of wave propagation (Russell and Hampson, 1991; Partyka et al., 1999). Features thinner than approximately one-quarter of the dominant wavelength are difficult to recover reliably, and any attempt to artificially increase resolution risks introducing noise or geological inconsistency without tight constraints.

Full Frequency Inversion (FFI) addresses these limitations from a different perspective. FFI is a seismic waveform-driven approach that exploits the lateral similarity of seismic trace shapes to guide the propagation of high-frequency impedance detail from well logs across the seismic volume (Liu et al., 2020). Instead of relying solely on the limited frequency content of the recorded wavefield, FFI uses trace waveform shape as a proxy for lithologic similarity, enabling the lateral distribution of well-measured high-frequency components in a data-driven, facies-aware manner. The outcome is an impedance volume that retains the low- and mid-frequency information necessary for regional structural and stratigraphic context, while also carrying high-frequency detail tied to geological control at the well scale.

For interpreters, the difference is tangible. Boundaries sharpen, thin beds emerge from composite responses, and lateral continuity of reservoir bodies becomes clearer and more geologically consistent. This

enhancement is particularly valuable in stratigraphic intervals where the vertical succession of lithologies is complex and where thin, laterally variable units exert significant control on production performance.

While the present study focuses on a clastic reservoir in the Browse Basin, the challenges it addresses—thin layering, subtle stratigraphic pinchouts, lateral heterogeneity—are common across many hydrocarbon provinces. This includes carbonate and mixed lithology settings, where thin high-energy shoals, interbedded tight facies, and diagenetic heterogeneity often fall below the vertical resolution of standard inversion approaches. By illustrating the application of FFI in a well-constrained clastic example, the workflow and benefits demonstrated here can inform similar applications in a wide range of geological contexts.

Geological Setting and Dataset

Regional Geological Framework

The Browse Basin is a prolific hydrocarbon province located on the Australian North-West Shelf. Structurally, it forms part of a northwest-trending extensional depocentre that developed during the protracted breakup of Gondwana from the Late Paleozoic through the Mesozoic. Multiple rifting episodes, separated by periods of relative tectonic quiescence, have created a complex basement architecture of horsts, grabens, and fault-bounded terraces (Keep et al., 2007). This tectonic template exerted a primary control on sediment pathways, accommodation space, and depositional patterns throughout the basin's history.

Syn-rift phases were dominated by fluvial and deltaic systems that prograded into fault-bounded depocentres. As rifting waned and the basin transitioned into a post-rift thermal subsidence regime, depositional systems shifted toward more open marine and shelf environments. This created a stratigraphic record in which coarser-grained, higher-energy intervals are often encased within finer-grained marine successions. The interplay of tectonics and sediment supply resulted in laterally variable sandbody distribution, often complicated by syndepositional faulting and localized accommodation changes.

The basin's petroleum system is sourced primarily from organic-rich marine shales of the Lower to Middle Jurassic, with migration into structural and stratigraphic traps within the Plover, Montara, and other key formations. Large gas and condensate accumulations are found in fields such as Ichthys, Crux, and Poseidon, confirming the effectiveness of this petroleum system (Paumard et al., 2014).

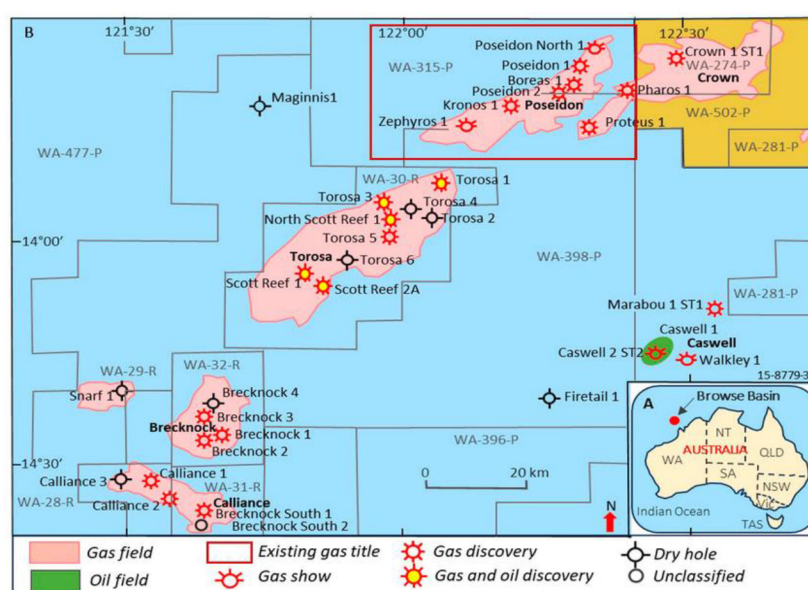


Figure 1—Location map of the Poseidon 3D seismic survey within the Browse Basin, offshore North-West Australia.

Plover Formation Stratigraphy and Depositional Setting

The Plover Formation, deposited during the Middle to Late Jurassic, is a mixed fluvio-deltaic to shallow marine succession. It typically comprises stacked, coarsening-upward parasequences of fine- to medium-grained quartzose sandstones interbedded with siltstones and shales. These sandstones commonly display sedimentary structures such as cross-bedding, ripple lamination, and bioturbation, reflecting variations in depositional energy. In places, tidal influence is indicated by inclined heterolithic stratification.

Reservoir quality in the Plover Formation varies considerably. Primary porosity is modified by diagenetic processes, including quartz overgrowths, localized carbonate cementation, and clay authigenesis. Clay content varies both vertically and laterally, influencing impedance contrasts and petrophysical properties. The thin nature of many sand bodies—ranging from less than a metre to ~15 m thick—means that a significant proportion of reservoir-quality intervals fall at or below the tuning thickness of the available seismic data. This presents a challenge for conventional inversion methods, which are unable to fully resolve these features without aliasing or amalgamation.

The distribution of thin shale drapes within the Plover is equally important. Even centimetre- to decimetre-scale low-permeability layers can act as baffles to vertical fluid flow, influencing production and sweep patterns. Their accurate mapping can materially impact reservoir simulation outcomes and field development planning.

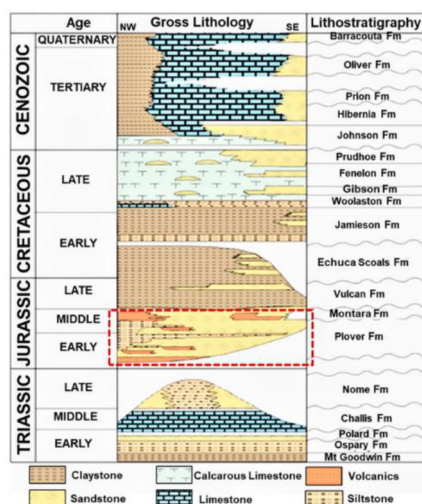


Figure 2—Generalized stratigraphic column of the Browse Basin showing the position of the Plover Formation and key overlying and underlying units.

Available Data

The dataset for this study comprises a publicly available 3D post-stack seismic volume over the Poseidon area and multiple exploration wells with comprehensive log suites and checkshot surveys. The seismic survey covers approximately 650 km² and was acquired with a marine streamer configuration, a nominal fold of 60, and a bin size of 12.5 m crossline by 25 m inline. The record length of 8 seconds two-way time ensures imaging of both reservoir and deeper stratigraphic intervals.

The dominant frequency in the Plover Formation interval is between 35 and 45 Hz, corresponding to a pre-inversion vertical resolution on the order of 12–15 m. This limitation means that beds thinner than this—especially those below ~3–4 m—are not individually resolvable using standard post-stack inversion.

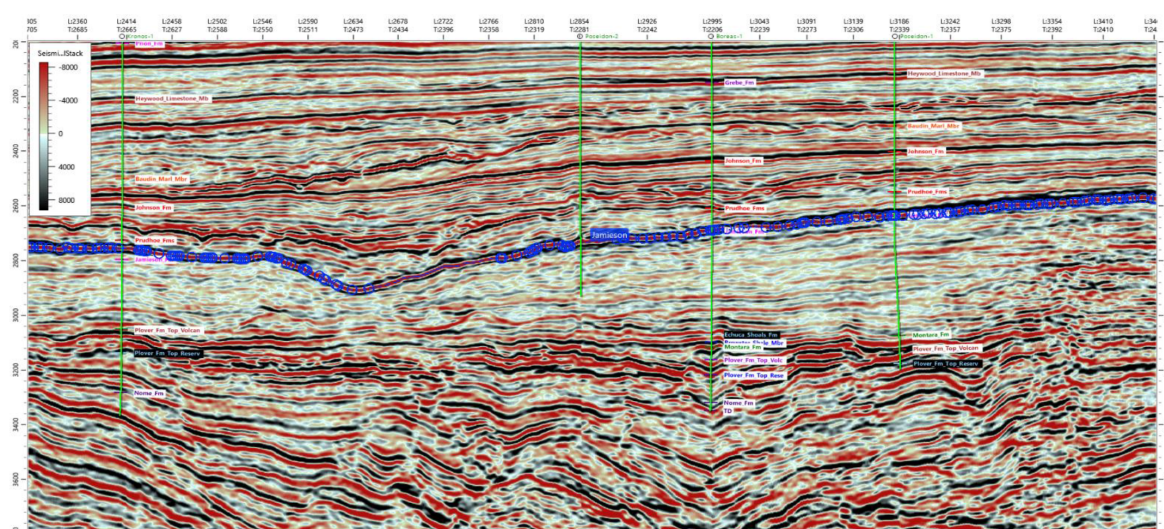


Figure 3—Arbitrary seismic line through the Poseidon 3D dataset showing the Plover Formation interval (blue horizon is Jamieson horizon picked by AI-driven technology).

Well data include gamma ray, resistivity, sonic, and density logs, enabling the derivation of acoustic impedance logs for calibration. Checkshot surveys provide accurate time-depth conversion and reduce misties between well and seismic domains. Horizons corresponding to the top Plover and other key stratigraphic boundaries (e.g., Jamieson Formation) were interpreted and used to constrain the inversion framework.

Data Conditioning

Amplitude-preserving seismic conditioning is critical for the success of waveform-driven inversion techniques. The Poseidon dataset underwent spectral balancing to compensate for attenuation, residual statics and phase rotation corrections to refine well ties, and targeted noise suppression to remove random and coherent noise without distorting true amplitude responses. Wavelet extraction at well locations ensured phase consistency across the inversion volume. These steps are essential to maintain the fidelity of waveform shapes, which form the basis of FFI's clustering and frequency extension process.

Methodology

Full Frequency Inversion (FFI) is a seismic waveform-driven inversion technique designed to extend the usable frequency bandwidth of post-stack inversion by integrating well-derived high-frequency detail into the seismic domain through a trace-shape similarity approach. In essence, it leverages the inherent lateral consistency of seismic waveform signatures as a proxy for lithologic association and uses this to propagate well-scale information across the seismic volume. The process is deterministic, data-guided, and anchored in physical relationships between waveforms and geology.

Well Data Preparation and Calibration

The workflow begins with rigorous well data preparation to ensure the reliability of the high-frequency impedance detail to be propagated. Sonic and density logs are first edited to remove spurious points, reconciled across overlapping tool runs, and upsampled to the seismic sample rate to prevent aliasing during integration. Checkshot data are used to establish precise time-depth relationships, minimizing timing mismatches in the well-to-seismic tie. Synthetic seismograms are generated for each well to verify phase and amplitude alignment with the conditioned seismic data.

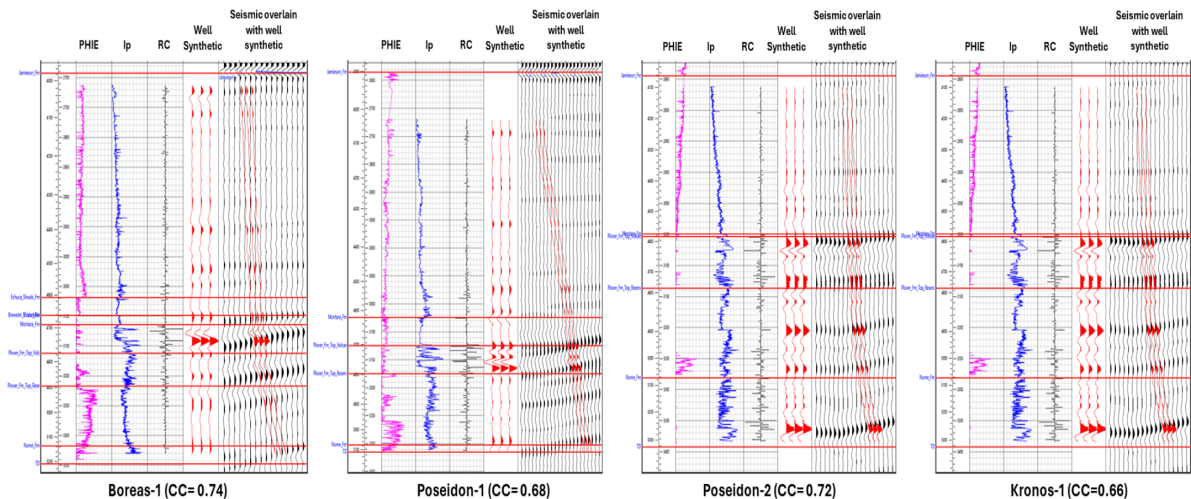


Figure 4—Example well tie from the Poseidon field showing Porosity, P-impedance, porosity, synthetic seismogram, and full-stack seismic response, with key formation tops marked.

Wavelet extraction is performed at each well location, either for the full inversion interval or for individual stratigraphic segments where phase characteristics vary. The extracted wavelet must represent the true seismic signature at that location and be free from processing artifacts.

Seismic Trace Analysis and Waveform Clustering

Once well data are calibrated, the inversion interval in the seismic volume is subjected to waveform similarity analysis. Each seismic trace is treated as a waveform, characterized by its amplitude, phase, and overall shape. These waveforms are compared and clustered into families that share similar shapes, under the principle that similar waveforms are likely sampling similar lithologic successions (Liu et al., 2020).

This clustering is performed without reliance on pre-defined facies models or interpreter bias. Instead, the clustering is purely data-driven, allowing geological associations to emerge organically from the seismic character. The spatial distribution of waveform families often reveals depositional trends, stratigraphic compartmentalization, or subtle facies transitions that may not be evident from amplitude alone.

Separation and Propagation of High-Frequency Detail

In parallel with seismic clustering, acoustic impedance logs are spectrally decomposed into three components:

- Low-frequency trend – representing the large-scale background impedance structure.
- High-frequency detail – containing thin-bed and small-scale lithologic variations measured at the well.
- Noise – incoherent components not linked to geology, which are removed prior to propagation.

The high-frequency detail component is then propagated laterally, but only along traces belonging to the same waveform family as the well tie at the corresponding depth. This constraint ensures that detailed impedance variations are only introduced where the seismic character supports a geological correlation, reducing the risk of spurious features.

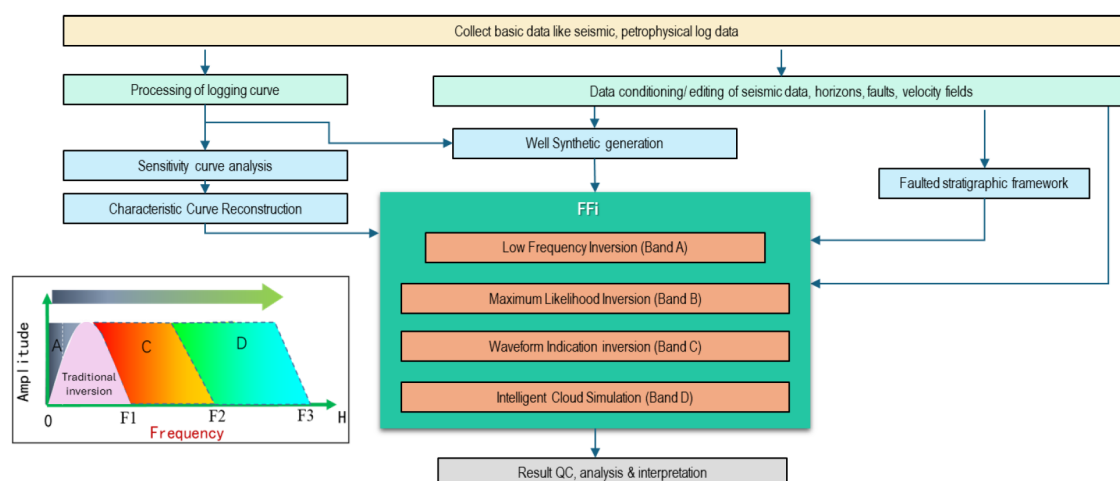


Figure 5—Full Frequency Inversion (FFI) workflow illustrating key stages from data preparation to inversion bands, frequency extension, and final quality control.

Inversion and Bandwidth Extension

The inversion itself is iterative, with each cycle balancing the fidelity of the seismic fit against the integration of high-frequency content. The initial inversion produces a model-based impedance consistent with the seismic and the low-frequency trend. The waveform-guided enrichment then injects the high-frequency detail, followed by a re-fit to the seismic to preserve amplitude and phase integrity.

The end result is an impedance volume that retains the low- and mid-frequency content of the seismic—critical for structural and regional stratigraphic mapping—while incorporating high-frequency components that sharpen stratigraphic boundaries, enhance thin-bed visibility, and improve lateral continuity of small-scale geological features.

Comparison to Conventional Inversion Approaches

Unlike conventional post-stack model-based inversion, which is limited to seismic bandwidth, or stochastic/geostatistical approaches, which may introduce uncontrolled variability, FFI uses deterministic, data-driven constraints. It combines the seismic's spatial coverage with the log's vertical resolution in a geologically consistent way. The result is a high-resolution impedance volume that can be directly integrated into facies mapping, net-to-gross estimation, and reservoir modeling.

Results

Application of Full Frequency Inversion (FFI) to the Plover interval in the Poseidon 3D seismic survey yielded an impedance volume that is qualitatively distinct from what would be expected from a conventional post-stack inversion. While both approaches honor the low- to mid-frequency trends present in the seismic data, the FFI output contains additional high-frequency detail that materially changes the interpretive outcome.

Spectral diagnostics indicated that the FFI impedance carried coherent energy at frequencies higher than the dominant range of the input seismic. This bandwidth extension is not simply an artifact of sharpening but reflects the targeted lateral propagation of well-derived high-frequency components guided by waveform clustering. The spectral increase translates directly into visible geological detail: stratigraphic boundaries appear sharper, adjacent lithologies are more clearly separated, and thin beds that were previously amalgamated into composite seismic responses become distinguishable.

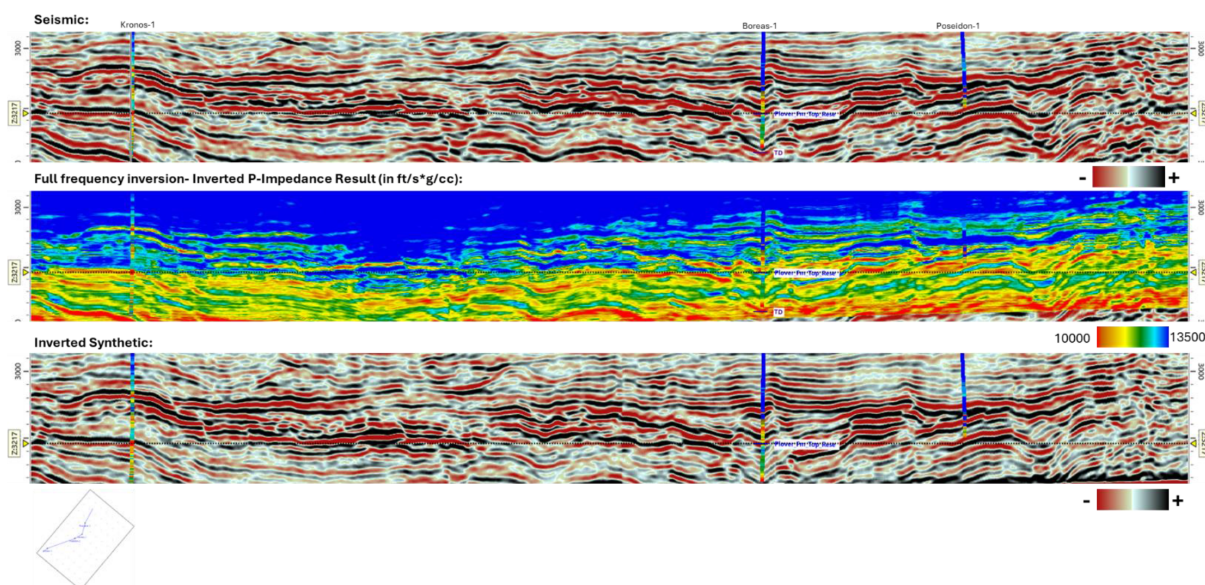


Figure 6—Seismic Full Frequency Inversion (FFI) Cross-Section Through the Plover Formation

At well locations, impedance traces from the FFI volume are expected to align more closely with measured log responses than those from conventional inversion. Thin coarsening-upward cycles, capping sands, and interbedded shales appear more faithfully in both their impedance magnitude and vertical position. Importantly, these improvements extend beyond calibration wells—blind well testing during workflow development showed that waveform clustering generalized well, allowing accurate high-frequency propagation into areas without direct well control.

Thin reservoir units within the Plover Formation, often less than 10 m thick, stand out more clearly in the FFI volume. These include isolated channel fills, laterally persistent mouth bars, and thin high-energy sand sheets that are critical to production planning. The improved vertical resolution also makes it easier to identify and map thin impermeable shale drapes that can act as baffles to vertical flow, influencing fluid movement during depletion or water injection.

Lateral continuity is also enhanced. Channel margins that previously appeared diffuse under tuning effects are now more sharply defined, allowing clearer delineation of channel widths and meander geometries. Pinchouts and subtle stratigraphic terminations, which can be key trapping mechanisms in certain intervals, are more confidently mapped. Fault offsets within the reservoir interval are also more distinct, enabling better estimation of throw and more reliable horizon correlation across faults.

From a workflow perspective, these qualitative improvements translate directly into more robust attribute analysis. When attributes such as relative impedance, sweetness, or derivative-based measures are extracted from the FFI volume, they inherit the enhanced resolution and sharper boundaries. This means that net-to-gross estimation benefits from clearer cutoffs, facies modeling gains more accurate input data, and volumetric calculations are grounded in a more geologically consistent impedance field.

The FFI results also integrate seamlessly into multidisciplinary interpretation workflows. For example:

Static modeling – The sharper impedance contrasts can be used to refine facies probability volumes and improve the conditioning of geomodel grids.

Dynamic modeling – Identification of baffles and barriers informs history-matching strategies and helps predict sweep efficiency during waterflood or gas injection.

Exploration screening – In frontier areas, the ability to resolve thin stratigraphic terminations can improve risk assessment for subtle stratigraphic traps.

While the present study centers on the Plover Formation, the qualitative nature of these improvements suggests that similar resolution gains would be expected in other geological settings with thin, laterally variable units—whether in clastic, carbonate, or mixed lithology contexts.

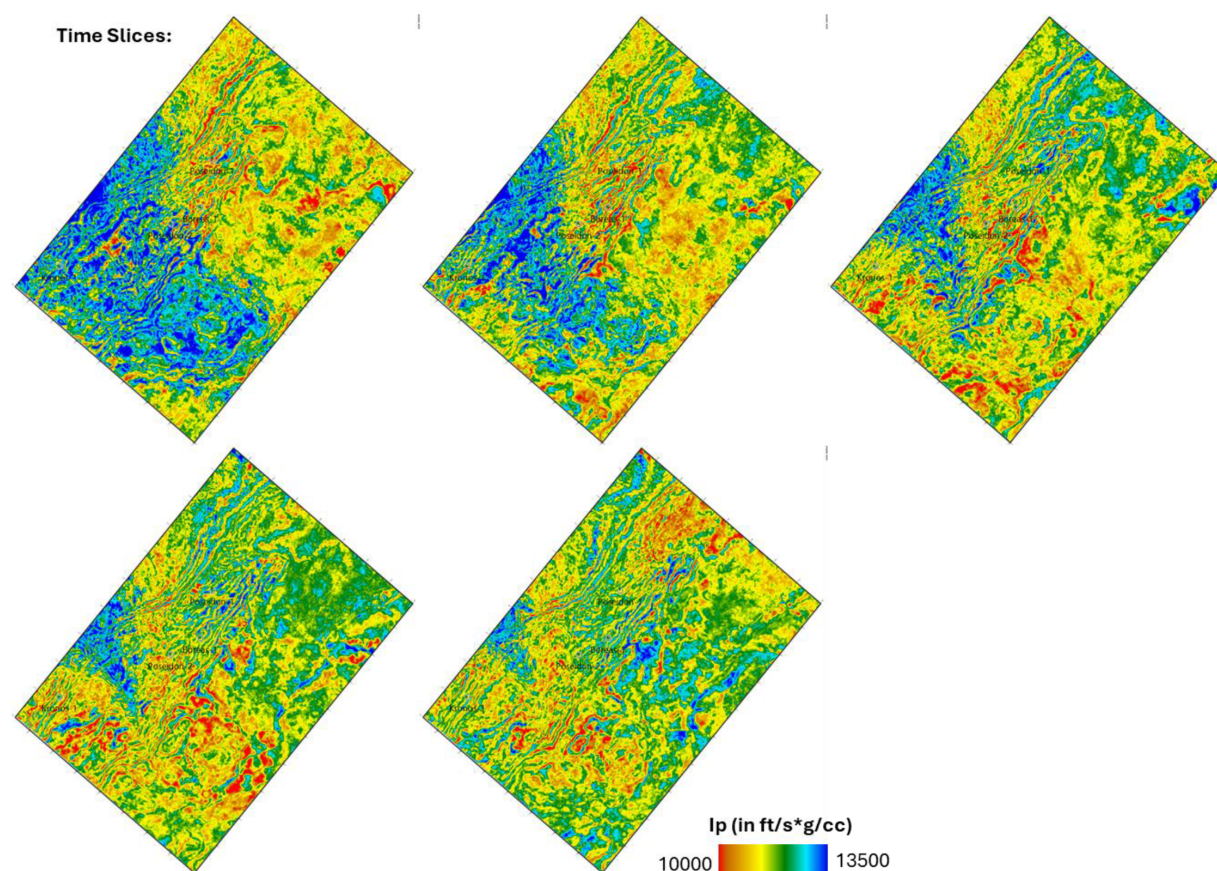


Figure 7—Full Frequency Inversion (FFI) inverted P-impedance seismic inversion time Slices across Plover formation.

Discussion

The results of this study underscore the potential of Full Frequency Inversion (FFI) to materially improve the interpretive value of seismic inversion products in reservoir characterization. By integrating waveform clustering with well-derived high-frequency impedance detail, FFI provides a means to overcome one of the most persistent limitations in conventional inversion—the inability to resolve geological features below the seismic tuning threshold.

From an operational standpoint, the implications are significant. In the Plover Formation, thin but laterally continuous sand bodies, as well as thin shale layers that act as flow barriers, can exert strong influence on fluid movement and recovery efficiency. When these features are blended into composite impedance responses, the resulting reservoir models tend to overestimate connectivity, underestimate compartmentalization, and misrepresent sweep behavior. The FFI-derived impedance volume mitigates these risks by providing a clearer and more geologically consistent picture of thin-bed distribution and lateral continuity.

Improved resolution in the inversion domain directly benefits static reservoir modeling. Facies probability volumes generated from FFI impedance are likely to be better constrained spatially, enabling more accurate population of geomodel grids. This, in turn, enhances the reliability of volumetric estimates and supports the design of well trajectories that target the most productive parts of the reservoir while avoiding non-reservoir or low-quality intervals.

In dynamic modeling, the enhanced detail helps identify potential baffles and barriers that may control sweep patterns during secondary recovery. For example, thin shale drapes mapped from the FFI volume can be incorporated into simulation models to test their impact on water breakthrough timing or gas migration pathways. This level of detail can refine field development strategies, influencing the placement of injectors, producers, and infill wells.

Beyond the specific case of the Plover Formation, the FFI methodology is applicable to a wide range of geological settings. In carbonate reservoirs, thin high-energy shoal deposits interbedded with lower-energy facies often play a similar role to thin clastic sand bodies, influencing connectivity and flow. In mixed carbonate–clastic systems, heterolithic bedding and tidal rhythmites present similar resolution challenges. The data-driven, waveform-guided approach of FFI is well suited to these contexts, provided that sufficient well control exists to supply high-frequency impedance information for propagation.

Integration into multidisciplinary workflows is straightforward. FFI volumes can be used directly in structural and stratigraphic interpretation, as inputs to rock physics analysis, or as conditioning data for machine learning models designed to classify facies or predict reservoir properties. This compatibility with existing tools and processes lowers the barrier to adoption and facilitates incremental implementation alongside more established inversion techniques.

While the benefits are clear, it is also important to recognize the methodological safeguards that underpin the quality of FFI results. Propagating high-frequency detail beyond the seismic bandwidth carries the risk of introducing artifacts if not carefully constrained. In this workflow, two guardrails are essential:

1. **Waveform clustering constraint** – High-frequency detail is only propagated along traces belonging to the same waveform family as the calibration well at the relevant depth, reducing the risk of geological inconsistency.
2. **Comprehensive quality control** – Spectral diagnostics, well tie validation, blind testing, and visual inspection ensure that the enhanced resolution is stratigraphically coherent and not the product of noise or acquisition footprint.

By adhering to these safeguards, FFI achieves a balance between extending resolution and maintaining geological plausibility. This balance is critical for interpreter confidence, particularly when inversion results will be used to make high-value decisions in field development.

In summary, the application of FFI in this study demonstrates a scalable, data-driven enhancement to conventional inversion workflows that is directly relevant to improving reservoir characterization. The methodology's adaptability to different depositional environments and its ability to integrate into multidisciplinary projects suggest it has broad applicability well beyond the current case study.

Conclusions

Full Frequency Inversion (FFI) applied to the Poseidon 3D seismic data over the Plover Formation produced an impedance volume that preserves seismic wavefield characteristics while incorporating well-scale resolution. The workflow's foundation is its waveform clustering approach, which recognises that traces with similar seismic shapes are likely sampling similar lithologic successions. By propagating high-frequency impedance detail from well logs along these waveform families, FFI extends inversion resolution beyond the limits set by the recorded seismic bandwidth.

This is a departure from traditional inversion methods, which are fundamentally constrained by the frequency content of the seismic wavelet. Conventional approaches can fit the available data faithfully but struggle to recover detail thinner than the seismic tuning thickness without resorting to stochastic variability or spectral shaping, both of which can introduce uncertainty. FFI breaks this constraint by introducing a deterministic, geologically guided high-frequency enrichment step, ensuring that additional detail is not random but spatially coherent and stratigraphically plausible.

In the Plover Formation, this novel approach enhanced thin-bed visibility, sharpened stratigraphic boundaries, and improved mapping of lateral continuity. The improvements directly support more accurate facies modeling, net-to-gross estimation, and the identification of small-scale flow barriers in reservoir studies. Although demonstrated here in a clastic setting, the method's principles are equally applicable to carbonate, mixed lithology, or other complex depositional systems where fine-scale heterogeneity governs reservoir performance.

Built-in safeguards—including waveform clustering constraints, rigorous well calibration, and comprehensive QC—ensure that the added resolution remains geologically valid. By bridging the gap between seismic-scale and well-scale information, FFI represents a significant step beyond the capabilities of conventional inversion and offers interpreters a richer, more actionable dataset for subsurface decision-making.

Acknowledgement

The authors thank the management of China National Petroleum Corporation Research and Development (CNPCRD) for supporting this study and granting permission to publish. We also acknowledge Geoscience Australia and the open data initiative by the Society of Exploration Geophysicists (SEG) for making the Poseidon 3D seismic dataset and related well data available to the public.

References

- Russell, B., and Hampson, D., 1991, The old and the new in seismic inversion: *The Leading Edge*, **10**(5), 39–43. <https://doi.org/10.1190/1.1436817>
- Partyka, G., Gridley, J., and Lopez, J., 1999, Interpretational applications of spectral decomposition in reservoir characterization: *The Leading Edge*, **18**(3), 353–360. <https://doi.org/10.1190/1.1438281>
- Liu, J., Wang, Y., and Hu, H., 2020, Full Frequency Inversion: Theory, implementation, and applications: *SEG Technical Program Expanded Abstracts 2020*, 1555–1559. <https://doi.org/10.1190/segam2020-3426221.1>
- Keep, M., Moss, S., and Collins, A. S., 2007, Neoproterozoic to recent tectonic history of the North West Shelf, Australia: *AAPG Bulletin*, **91**(11), 1731–1754. <https://doi.org/10.1306/06200706146>
- Paumard, V., Bouroullec, R., and Chace, B., 2014, Seismic imaging and inversion for reservoir characterization in the Browse Basin, Australia: *SEG Technical Program Expanded Abstracts 2014*, 2463–2467. <https://doi.org/10.1190/segam2014-1502.1>